

Week 4 — Bayesian Generalization

Friday, May 22, 2026

Prof. Joseph Austerweil

Agenda

Welcome + Clusters walkthrough	0:00
The generalization problem	0:10
The Bayesian generalization framework + size principle	0:18
Rectangle game + number game	0:40
Break	1:08
Student presentation — Shohei	1:15
No Free Lunch	1:40
Hierarchical Bayes + close	1:50

Assignment 1 — Clusters

Clusters — what and when

Assignment 1: Gaussians, Categories, and Clusters

- Due **Fri Jun 5, 2026, 8:00 PM** — worth 7.5%
- **Problem 1** — Gaussian-Gaussian conjugate model → *you can do this today* (it's Week 3 material)
- **Problem 2** — Gaussian mixture / categorization → Bayes' rule over two category Gaussians
- **Problem 3** — clustering → leans on this week + T3 Ch 5

Clusters — which notebook

- `clusters.ipynb` is the canonical stencil — the GenJAX path
- `clusters_python.ipynb` and `clusters_nosoln.Rmd` — non-GenJAX paths, *same math, same credit*
- Matlab available on request
- “Open in Colab” links are live on the assignments page — one click, no local setup
- Read the **assignment PDF first** — it has the problem statements and all the math

Assignments page: hml.chibatech.dev/assignments.html

Where we are

Welcome + Clusters walkthrough	0:00
The generalization problem	0:10
The Bayesian generalization framework + size principle	0:18
Rectangle game + number game	0:40
Break	1:08
Student presentation — Shohei	1:15
No Free Lunch	1:40
Hierarchical Bayes + close	1:50

The generalization problem

Chibany's lunches

Chibany has had three tonkatsu lunches this week. Each weighed about **500 g**.

Today a lunch arrives weighing **480 g**. Is it tonkatsu?

What about **700 g**? What about **350 g**?

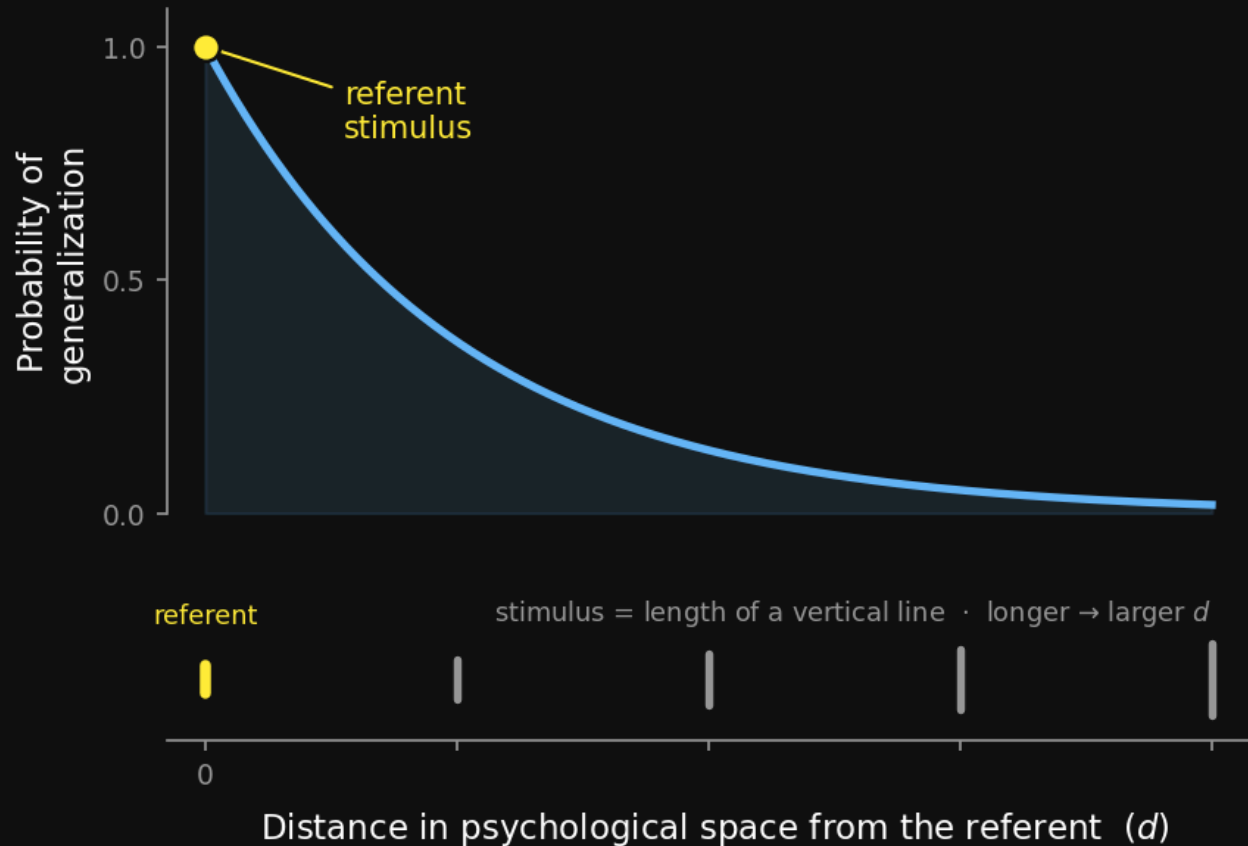
What just happened

Generalization — deciding when to extend a property from observed examples to a *novel* stimulus.

- No two stimuli are ever identical → you *must* generalize to act at all
- It is everywhere: word learning, categorization, object recognition, property induction, stereotypes
- It is the core problem of inductive inference

Shepard's universal law

Shepard (1987) — across species and domains, the probability of generalization **decays exponentially** with distance in **psychological space** (*the perceived distance, after the mind represents the stimulus — not raw physical space*).



One law, one equation:

$$g(d) = e^{-d}$$

- g — probability of generalization
- d — distance in psychological space from the referent

The same curve holds across **species, senses, and stimulus types** — pitch, colour, line length, faces. Shepard called it *universal* for that reason.

But the law is **descriptive**: it says generalization decays exponentially, not *why*. Today's Bayesian framework will **derive** that exponential — it falls out of the posterior.

Poll — Shepard's universal law

Shepard's universal law of generalization says generalization ...

- **A.** decays exponentially in psychological space
- **B.** decays exponentially in stimulus space
- **C.** grows exponentially in psychological space
- **D.** grows exponentially in stimulus space

Poll — answer

A. Decays exponentially in psychological space.

“Stimulus space” is the trap — generalization isn’t governed by physical distance but by *perceived* distance. A model of generalization therefore needs a model of the **psychological space**. That is exactly what the Bayesian framework supplies next.

Where we are

Welcome + Clusters walkthrough	0:00
The generalization problem	0:10
The Bayesian generalization framework + size principle	0:18
Rectangle game + number game	0:40
Break	1:08
Student presentation — Shohei	1:15
No Free Lunch	1:40
Hierarchical Bayes + close	1:50

The Bayesian generalization framework

The idea — concepts as hypotheses

Instead of measuring distance directly, posit a **space of candidate concepts** and let Bayes do the generalizing.

- A concept = a **set of stimuli** that share the property
- Shepard's term: a **consequential subset** — the subset of things the property “applies to”
- Generalization becomes: *which concepts are consistent with the examples, and do they contain the new stimulus?*

Notation lock-in

- h — a **hypothesis**: one candidate concept, i.e. a *set* of stimuli
- $\mathcal{H}$ — the **hypothesis space**: all candidate h
- $X = \{x_1, \dots, x_n\}$ — the observed **examples** of the concept
- y — a **novel stimulus** we must judge
- C — the (unknown) true concept

The three ingredients

Prior $p(h)$ — domain knowledge: which concepts are *natural* before any data.

Likelihood $p(X \mid h)$ — how probable the examples are if h is the true concept.

Posterior $p(h \mid X)$ — belief in h after seeing the examples:

$$p(h \mid X) \propto p(X \mid h) p(h)$$

The hypothesis space IS a prior

Choosing $\mathcal{H}$ is already a **strong prior**.

Any concept not in $\mathcal{H}$ has $p(h) = 0$ — it can never be learned, no matter the data.

So: a learner's inductive bias lives in *which hypotheses it even considers*.

Generalization = a posterior-weighted vote

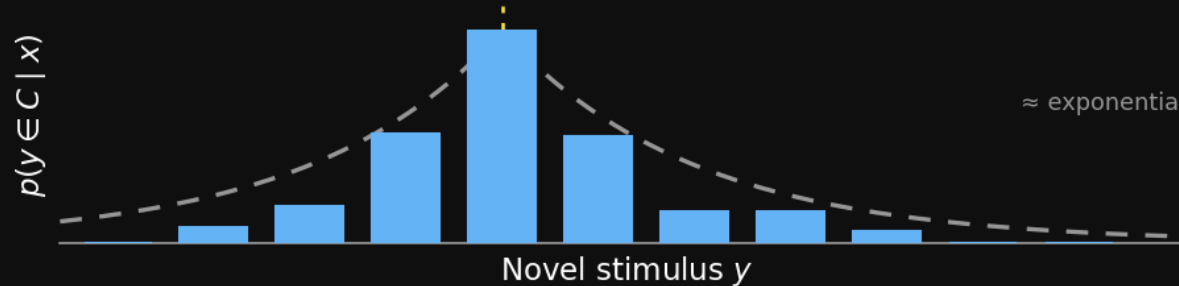
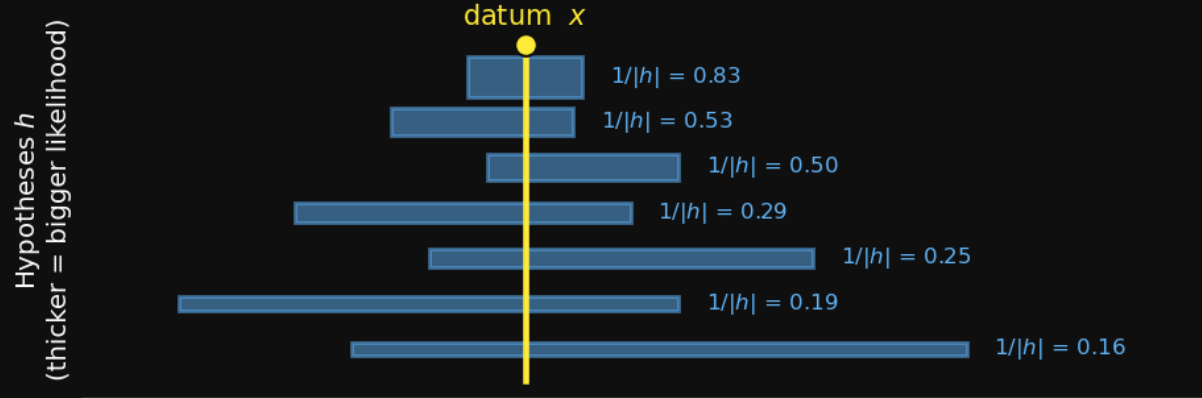
Probability that the novel stimulus y has the property:

$$p(y \in C \mid X) = \sum_{h \in \mathcal{H}} \mathbf{1}[y \in h] p(h \mid X)$$

Every hypothesis votes. Its vote is **its posterior weight**, and it votes “yes” only if it *contains* y .

$\mathbf{1}[y \in h]$ — the indicator: 1 if y is in the set h , else 0.

One datum → the posterior-weighted vote



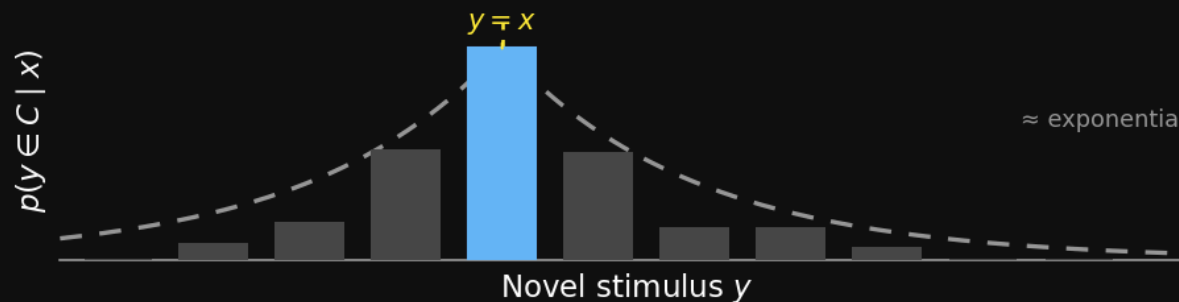
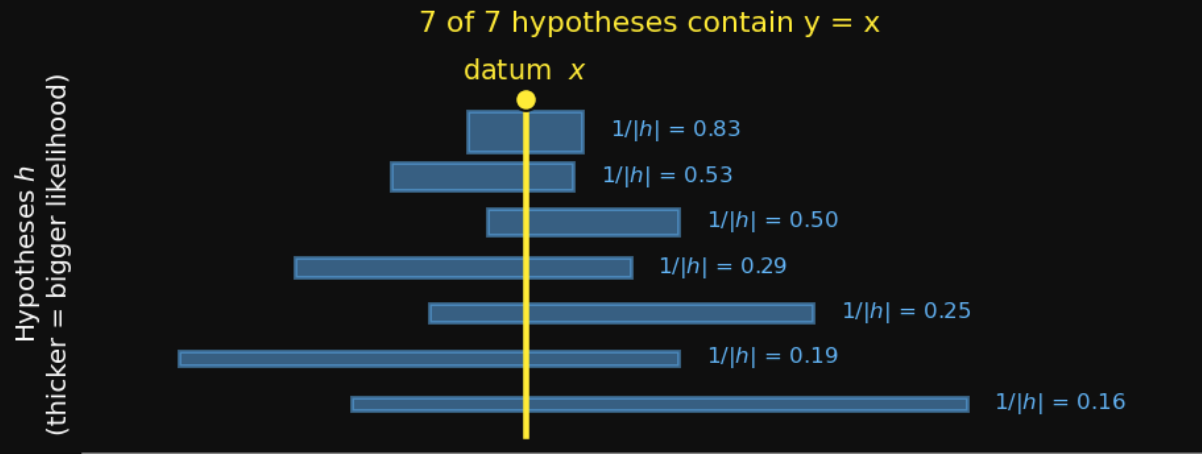
The hypotheses. Observe one datum x . Every interval that **contains** x is a live hypothesis h . A bar's thickness is its strong-sampling likelihood $1/|h|$ — a *smaller* interval is a *thicker* bar. Flat prior, so each h 's posterior weight $\propto$ its likelihood.

The vote. For each candidate y , add up the posterior of every h that contains it:

$$p(y \in C | x) = \sum_h \mathbf{1}[y \in h] p(h | x)$$

Next three slides: walk y outward — $x, x+1, x+2$ — and watch the vote shrink as hypotheses drop out.

Vote for $y = x$



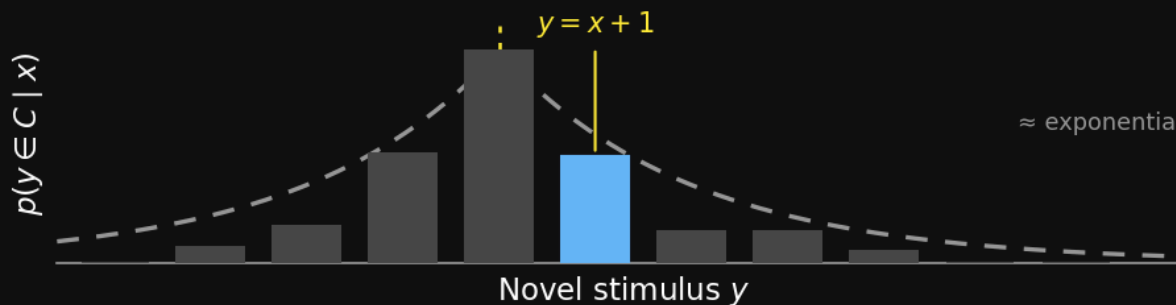
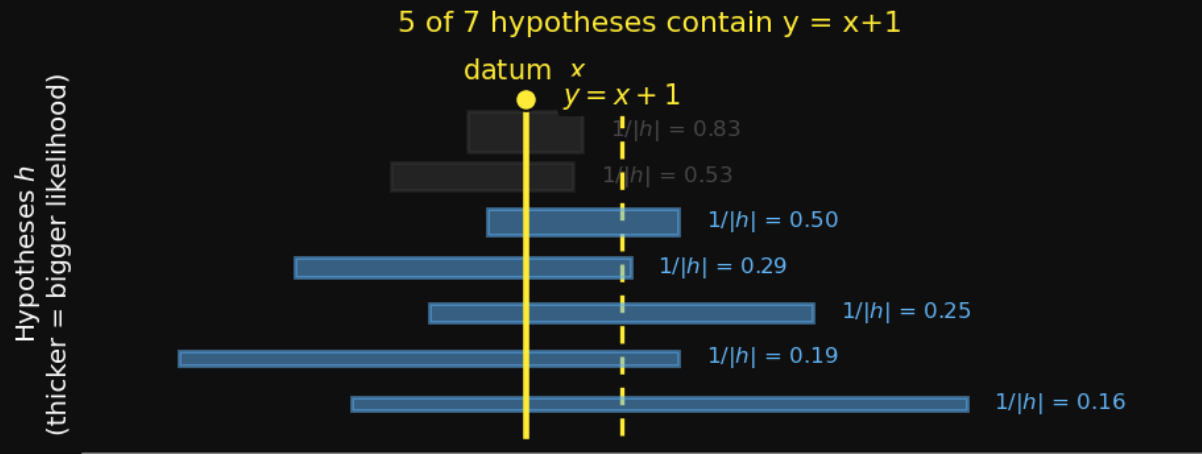
The question. Does the property generalize to $y = x$ — the observed datum itself?

Who votes. Every hypothesis in $\mathcal{H}$ was built to contain x , so **all 7 of 7 vote “yes”**. Not a single one drops out — there is no hypothesis that excludes the very point you observed.

The bar. The vote at x sums *all* the posterior weight — $\sum_h p(h | x) = 1$. This is the **tallest bar the gradient can have**.

The baseline. Generalization to the datum itself is near-certain — and it is the peak that every other y is measured against. As we step y away from x , hypotheses start dropping out, and the bar can only fall.

Vote for $y = x + 1$



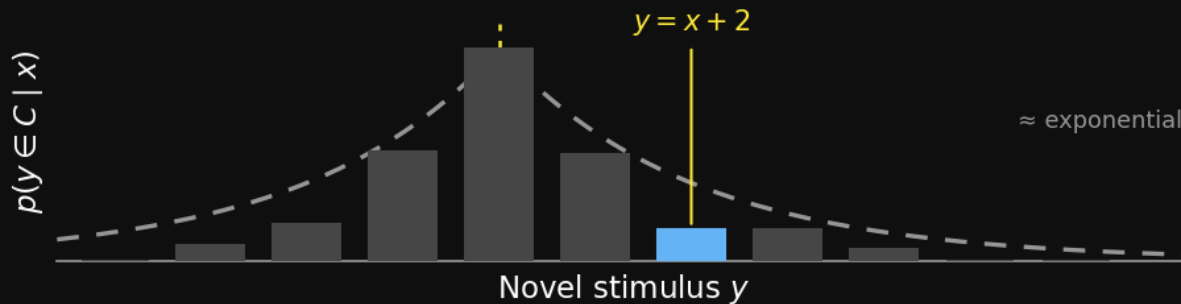
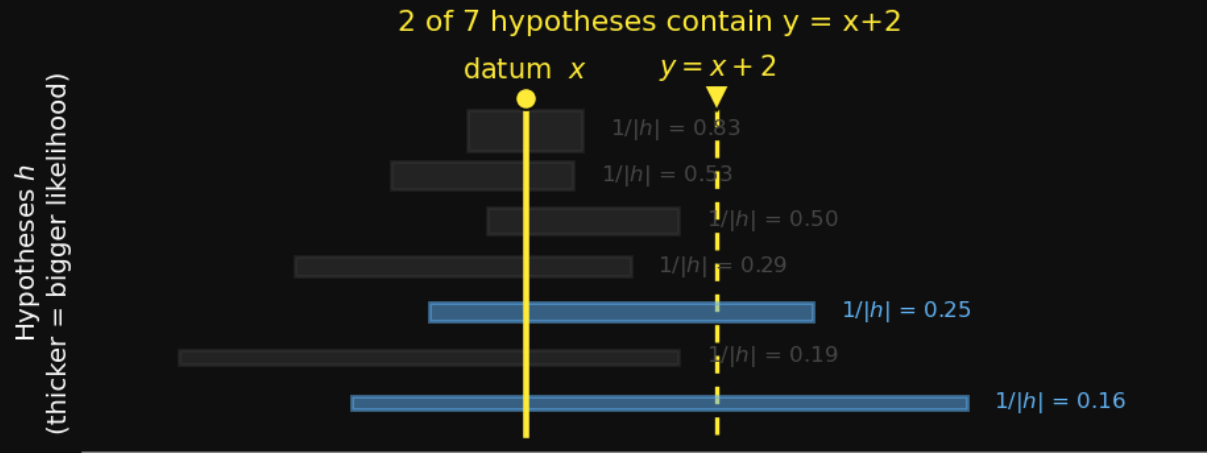
The question. Step one unit out from the datum: does the property generalize to $y = x + 1$?

Who drops. The two smallest intervals no longer reach y — y falls outside them, so they grey out and vote “no”.
5 of 7 still vote “yes”.

The bar. Fewer hypotheses contribute → a **shorter** bar than at x .

Why it falls fast. Note *which* hypotheses dropped: the **smallest** intervals — and by the size principle those carry the **most** posterior weight (thickest bars). Losing the heavy-weight votes first is exactly why the gradient decays steeply near x .

Vote for $y = x + 2$



The question. Two steps out: does the property generalize to $y = x + 2$?

Who's left. Only the 2 widest intervals still reach y — and those are the *least* likely (thinnest bars). **2 of 7 vote.**

The bar. Few votes, and only low-weight ones → the bar is short.

The payoff. Sweep y across every point and the bar heights trace an **approximately exponential decay** — Shepard's universal law, **derived** from the model, not assumed.

The size principle

Where did the examples come from?

The likelihood $p(X \mid h)$ depends on *how you assume the examples were generated*. Two assumptions:

Weak sampling — examples generated some other way, then *labeled* by whether they fall in h .

Strong sampling — each example drawn *uniformly at random* from *within* h .

The two likelihoods

Weak sampling

$$p(X \mid h) = \begin{cases} 1 & \text{all } x_i \in h \\ 0 & \text{else} \end{cases}$$

Size-blind: a hypothesis either contains the data or it doesn't. **Does not depend on $|h|$.**

$|h|$ — the size of hypothesis h (how many stimuli it contains).

n — number of examples.

Strong sampling

$$p(x \mid h) = \frac{1}{|h|} \Rightarrow p(X \mid h) = \left(\frac{1}{|h|} \right)^n$$

Each example drawn *uniformly from inside h* . **Smaller $|h| \rightarrow$ higher likelihood**, exponentially so in n .

The size principle

Under strong sampling: **smaller hypotheses get higher likelihood** — and *exponentially* more so as the number of examples n grows.

$$p(X \mid h) = \left(\frac{1}{|h|} \right)^n$$

So a small $|h|$ wins — and **wins faster with more data**, because the exponent n magnifies any size advantage.

The mechanism behind both games today.

Why — the suspicious coincidence

You see the examples {60, 80, 10, 30}.

- If the concept is “**multiples of 10**” — unremarkable, that’s just what its members look like.
- If the concept is “**even numbers**” — it’s a **suspicious coincidence** that not one of the four was 2, 4, 6, 8, ...

Strong sampling *penalizes* the big hypothesis for failing to predict the tight clustering you actually saw.

Poll — strong sampling

What is strong sampling?

- **A.** Each stimulus is generated uniformly at random from the true hypothesis
- **B.** A stimulus has probability one given the true hypothesis
- **C.** Larger hypotheses are given smaller prior probability
- **D.** Smaller hypotheses are given smaller prior probability

Poll — answer

A. Uniformly at random from the true hypothesis.

B describes *weak* sampling — membership gives likelihood 1, regardless of size. **C / D** describe a *prior* over hypotheses; the size principle is about the **likelihood**, not the prior. Strong sampling = “the examples were drawn *from inside* the concept” — and that is what makes size matter.

Where we are

Welcome + Clusters walkthrough	0:00
The generalization problem	0:10
The Bayesian generalization framework + size principle	0:18
Rectangle game + number game	0:40
Break	1:08
Student presentation — Shohei	1:15
No Free Lunch	1:40
Hierarchical Bayes + close	1:50

Rectangle game — continuous concepts

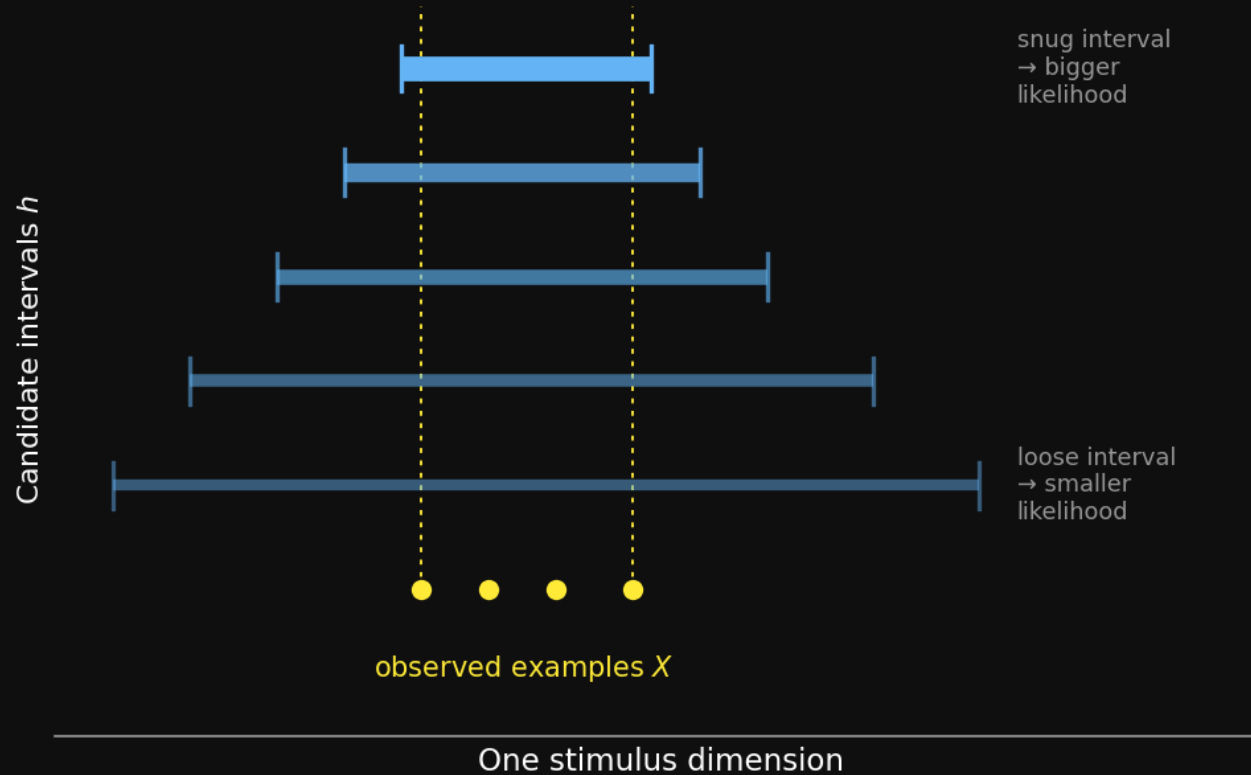
Gnarbles

A **gnarble** is a rectangle whose dimensions fall in some interval.

Tenenbaum's cover story: a **2-D concept** is an axis-aligned rectangle in a feature space — e.g. *healthy levels* of insulin × cholesterol.

You observe a few examples drawn from inside the rectangle. **Which other points are gnarbles?**

Start in 1-D



Concept = an **interval** $[\ell, u]$ on one dimension.

- $\mathcal{H} = \text{all intervals}$
- Strong sampling $\rightarrow \text{likelihood} \propto \left(\frac{1}{u - \ell}\right)^n$
- Hypothesis size = the **length** $u - \ell$

Every interval that contains the data is a hypothesis; the **snug** ones get the most posterior weight.

The generalization gradient

Plot $p(y \in C \mid X)$ as y moves along the dimension:

- **High** inside the range of the examples
- **Decays** as y moves outside that range

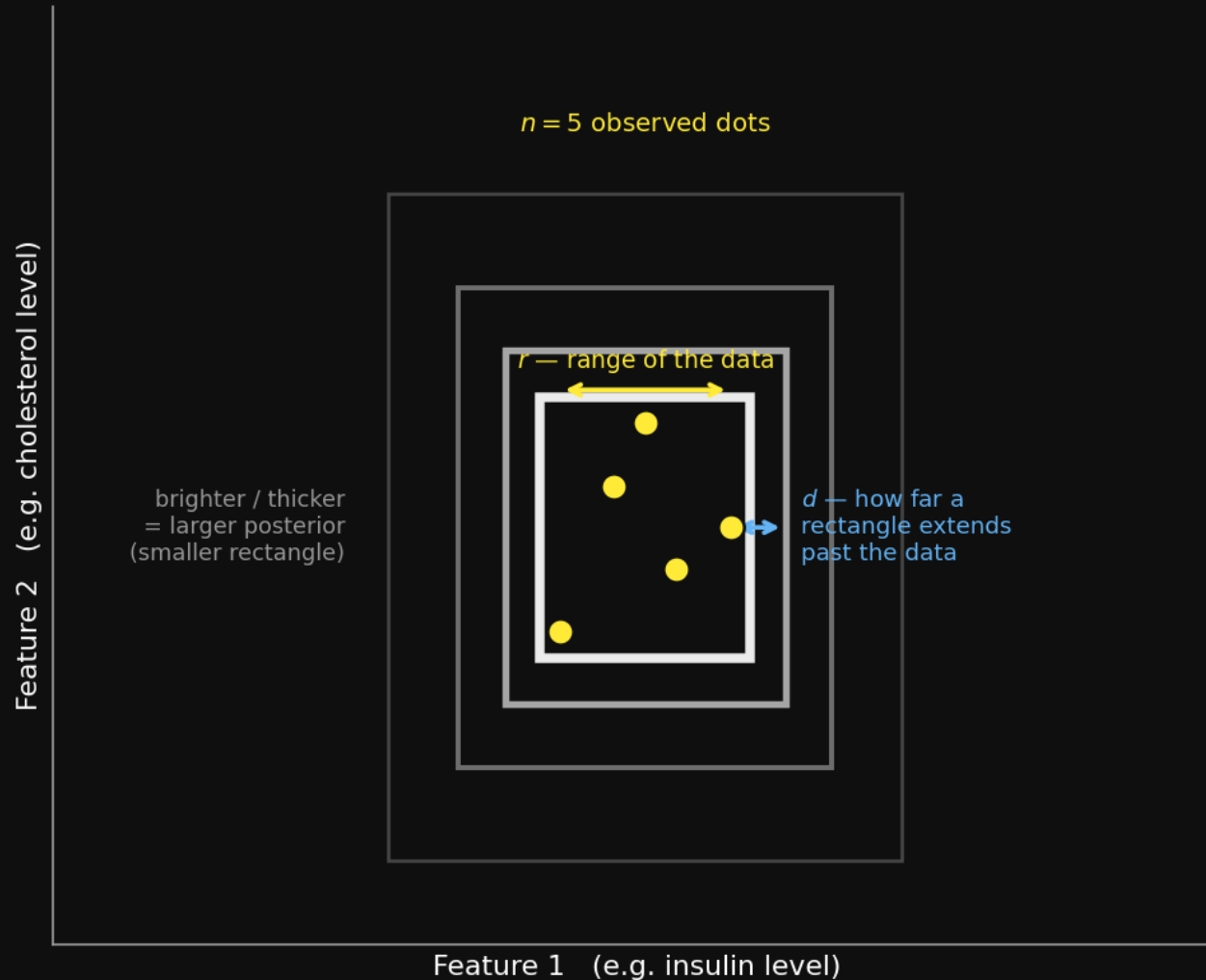
The decay is *exponential* — Shepard's universal law, now **derived** rather than assumed.

More examples → tighter generalization

- **One example** → broad, diffuse generalization (many interval sizes survive)
- **Many examples** → tight generalization, hugging the data range

Why: the size principle. With large n , big intervals lose likelihood *exponentially* fast — only the small, snug intervals keep posterior mass.

Into 2-D — the rectangle game



Same machinery, one dimension up: a concept is an axis-aligned **rectangle**.

- Observe n dots inside the true rectangle
- Every rectangle that **encloses all** n is a hypothesis
- Smaller rectangle $\rightarrow$ bigger likelihood (brighter / thicker)

r — the range the data spans. d — how far a rectangle extends past that range.

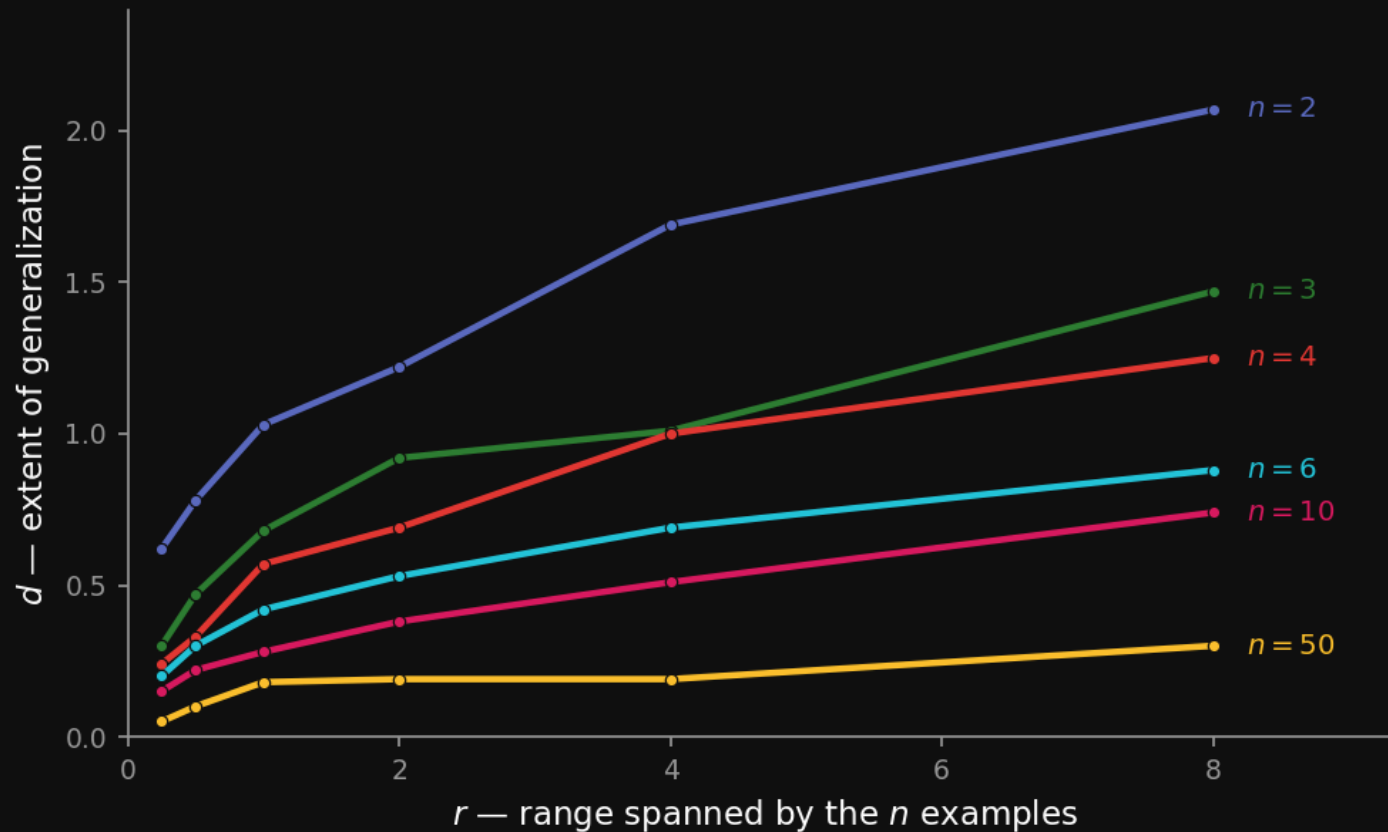
The rectangle experiment

Tenenbaum (1999) ran this as a behavioural experiment.

- On each trial, subjects saw n **dots** drawn from “an arbitrary rectangle of healthy insulin / cholesterol levels”
- They drew the rectangle they thought the dots came from
- n varied from **2 to 50**; the data range r varied across trials
- The measure: d — how far past the data range r the drawn rectangle extends

The result — d vs. r , by n

Average data, 6 subjects (Tenenbaum & Griffiths 2001)

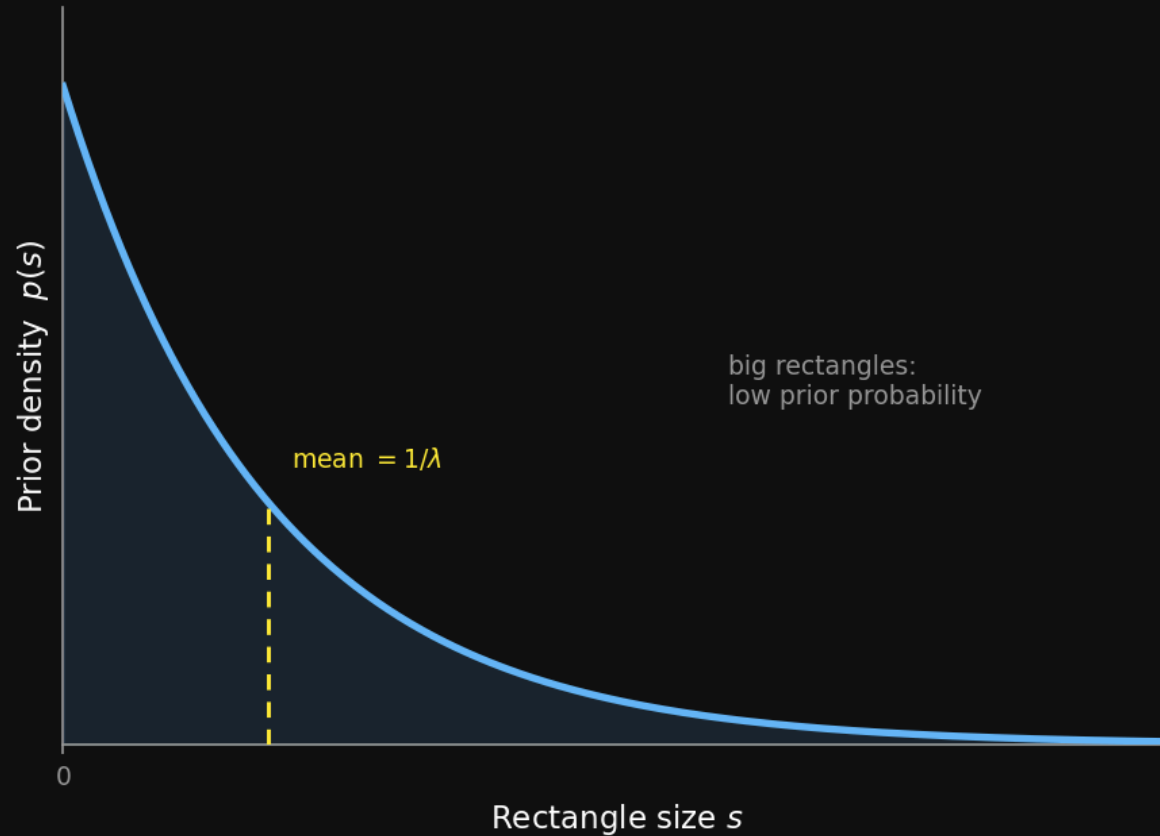


Each curve is one value of n .

- Fewer examples → generalize further ($n = 2$ on top)
- More examples → d collapses toward the data ($n = 50$ at the bottom)
- d grows gently with r

The size principle, measured in **human behaviour**: more data tightens the concept — and the Bayesian model reproduces every curve.

One fix — an exponential prior



The size principle alone slightly **over-extends** the rectangle. The fix: a prior that makes large rectangles less likely.

The exponential distribution — our first new distribution today. For a size $s \geq 0$:

$$p(s) = \lambda e^{-\lambda s}$$

- Always decreasing — **small s favoured**
- One parameter $\lambda > 0$; the mean is $1/\lambda$
- Larger $\lambda \rightarrow$ faster decay $\rightarrow$ stronger pull toward small rectangles

The rectangle game in one line

The continuous concept learner is **just the framework equation** with $\mathcal{H}$ = intervals / rectangles.

$$p(y \in C \mid X) = \sum_h \mathbf{1}[y \in h] p(h \mid X)$$

Nothing new — only the choice of $\mathcal{H}$.

Number game — discrete concepts

The number game

A simple task (Tenenbaum, 1999):

- I have a concept — a set of numbers between 1 and 100
- You see one or more “yes” examples
- You judge: is some other number a “yes”?

Watch what your own judgments do as examples come in.

What people actually do

Human generalization judgments (Tenenbaum's $N = 20$ subjects):

- Examples {60} → **diffuse** similarity: many numbers get moderate “yes”
- Examples {60, 80, 10, 30} → sharp “**multiples of 10**”
- Examples {60, 52, 57, 55} → sharp “**numbers near 60**”

One example → graded. Four examples → a crisp **rule**.

Two things to explain

1. Generalization can look **similarity-based (graded)** or **rule-based (all-or-none)** — and people switch between them.
2. People learn a concept from **just a few examples**.

One model — the Bayesian framework — produces *both*, with no extra machinery.

The discrete hypothesis space

$\mathcal{H}$ for the number game has two kinds of hypothesis:

Mathematical properties (~24) — even, odd, primes, squares, cubes, multiples of k , powers of k .

Magnitude intervals — “numbers in $[a, b]$ ”: e.g. 10–20, 30–45.

The **prior** $p(h)$ weights these families against each other.

Size principle, by the numbers

Two candidate concepts for numbers in 1–100:

- **Multiples of 2** — 50 numbers $\rightarrow p(x \mid h) = \frac{1}{50} = 2\%$ each
- **Multiples of 10** — 10 numbers $\rightarrow p(x \mid h) = \frac{1}{10} = 10\%$ each

One example: $x = 60$

$$p(60 \mid \text{mult-2}) = \frac{1}{50} \quad p(60 \mid \text{mult-10}) = \frac{1}{10}$$

Multiples of 10 is already **5×** more likely — but only 5×. With one example, many hypotheses stay in contention → **graded generalization**.

Four examples: {10, 30, 60, 80}

$$p(X \mid \text{mult-2}) = \left(\frac{1}{50}\right)^4 \approx 1.6 \times 10^{-7}$$

$$p(X \mid \text{mult-10}) = \left(\frac{1}{10}\right)^4 = 10^{-4}$$

Now multiples of 10 is **~625×** more likely. The 5× edge got raised to the 4th power → a crisp **rule**.

From likelihood to posterior

Likelihoods aren't beliefs yet. Take a **two-hypothesis** model — $\mathcal{H} = \{ \text{multiples of 10, even numbers} \}$, both containing every example — with a flat prior, and turn the crank:

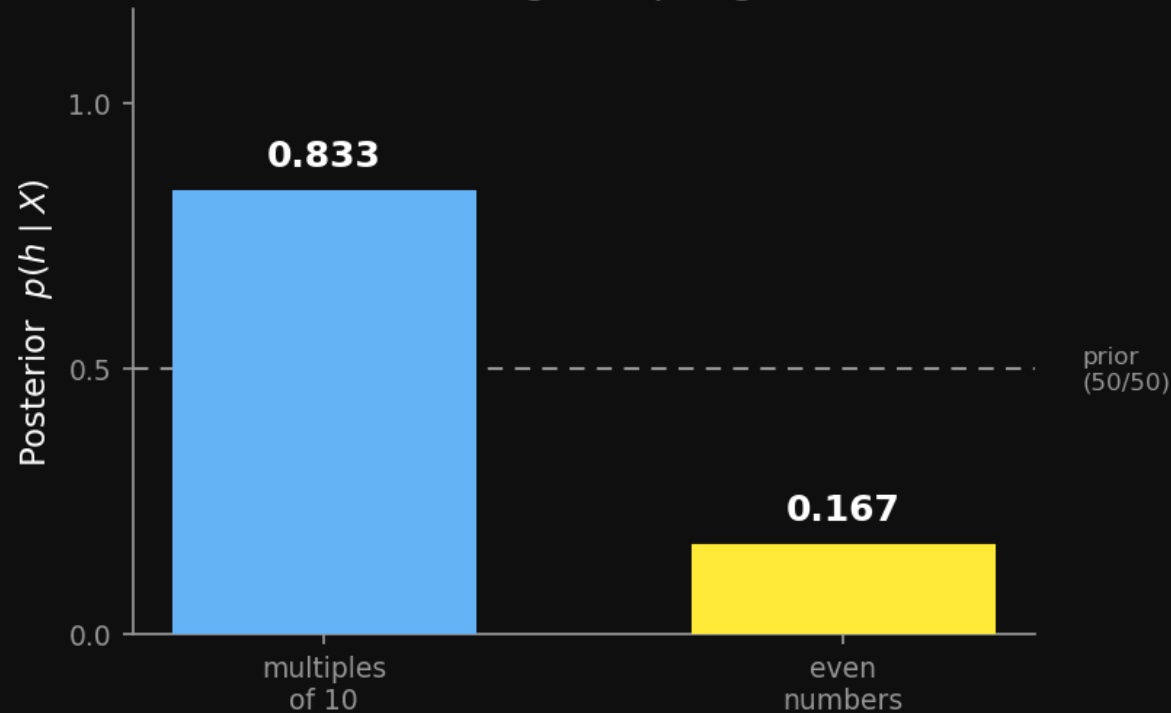
$$p(h \mid X) = \frac{p(X \mid h) p(h)}{\sum_{h'} p(X \mid h') p(h')}$$

Next: the posterior under **strong** vs. **weak** sampling, for $X = \{60\}$
then $X = \{60, 80, 10, 30\}$.

Strong sampling — one example

$X = \{60\}$ · one example

Strong sampling



The data: $X = \{60\}$ — a single example. Strong sampling:

$$p(h | X) \propto \left(\frac{1}{|h|}\right)^1 \cdot 0.5$$

- mult-10: likelihood $\frac{1}{10}$ · even: $\frac{1}{50}$
- Normalize the two → **0.83 vs. 0.17**

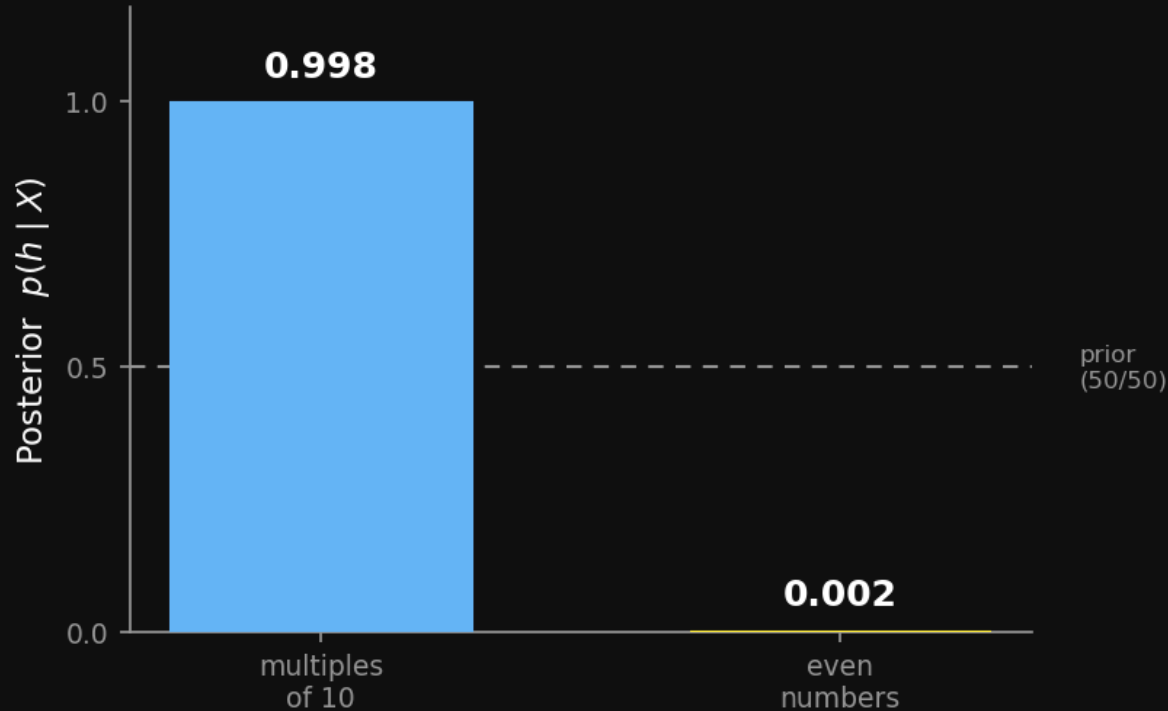
Read it: one example already **tilts** belief toward the smaller hypothesis — but only gently. A 5:1 likelihood ratio is not decisive, so plenty of posterior mass still sits on “even numbers”.

This is the **graded** regime — belief shifts, but no rule yet.

Strong sampling — four examples

$X = \{60, 80, 10, 30\}$ · four examples

Strong sampling



$X = \{60, 80, 10, 30\}$, strong sampling:

$$p(h | X) \propto \left(\frac{1}{|h|}\right)^4 \cdot 0.5$$

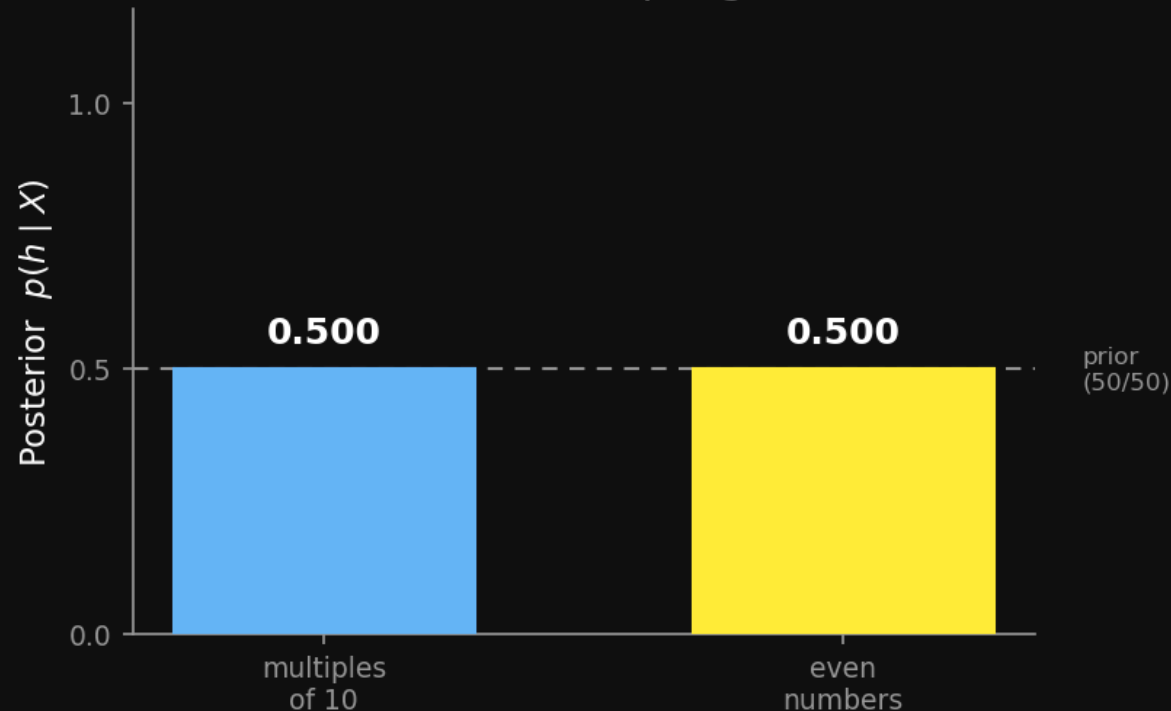
- The 5:1 ratio is now $5^4 = 625:1$
- Normalize → **0.998 vs. 0.002**

Four “even” numbers, none of them odd-looking — that’s the **suspicious coincidence**. Strong sampling all but rules “even” out.

Weak sampling — eliminate, but don't rank

both hypotheses survive $\rightarrow$ posterior = prior (0.5 / 0.5)

Weak sampling



Weak-sampling likelihood is 1 **for any h that contains the data**, 0 otherwise — *no $|h|$ dependence*.

What it can do: a datum *outside* h kills it — likelihood 0. Weak sampling **does** move the posterior, by **ruling hypotheses out**.

What it can't do: among the hypotheses that *all still contain the data*, it has no preference — every survivor keeps likelihood 1, so the posterior over them stays at the **prior**.

Here both hypotheses contain every example — nothing is ruled out:

$$p(h | X) = \frac{1 \cdot 0.5}{1 \cdot 0.5 + 1 \cdot 0.5} = 0.5$$

So weak sampling can't see the suspicious coincidence — it can't tell “multiples of 10” from “even numbers”.

The number game in one line

Same equation as the rectangle game — $\mathcal{H}$ is now **discrete**.

$$p(y \in C \mid X) = \sum_h \mathbf{1}[y \in h] p(h \mid X)$$

Graded vs. rule-like generalization both **fall out of the posterior** — no extra mechanism. The size principle's exponent does the switching.

Where we are

Welcome + Clusters walkthrough	0:00
The generalization problem	0:10
The Bayesian generalization framework + size principle	0:18
Rectangle game + number game	0:40
Break	1:08
Student presentation — Shohei	1:15
No Free Lunch	1:40
Hierarchical Bayes + close	1:50

Break

Bridge — Shohei's paper

Up next: Shohei presents **Tenenbaum & Xu (2000), *Word learning as Bayesian inference***.

A child hears “*this is a dax*” pointing at three Dalmatians. Is a poodle a dax? A cat?

This is the **number game**, with $\mathcal{H}$ = candidate word meanings (subordinate / basic-level / superordinate). The **size principle** explains why three subordinate examples → a subordinate meaning.

Watch for the size principle doing the work.

Student presentation — Shohei

Where we are

Welcome + Clusters walkthrough	0:00
The generalization problem	0:10
The Bayesian generalization framework + size principle	0:18
Rectangle game + number game	0:40
Break	1:08
Student presentation — Shohei	1:15
No Free Lunch	1:40
Hierarchical Bayes + close	1:50

No Free Lunch

The No Free Lunch theorem

Wolpert (1996) — averaged over all possible worlds, no learning algorithm beats any other.

Sequence prediction: you see x_1, x_2 , predict x_3 .

- If *every* hypothesis about the sequence is equally possible ...
- ... then for any data, every prediction is exactly **50/50**.

What NFL means for us

A learner only works because the distribution over worlds is **constrained** — i.e. because it has a **non-flat prior**.

- Generalization is *impossible* without inductive bias
- Recall: the hypothesis space $\mathcal{H}$ and the prior $p(h)$ from Block 3
- They are not bookkeeping — they are *the entire reason* learning is possible

Poll — No Free Lunch

What is the No Free Lunch theorem (for prediction)?

- **A.** When all hypotheses are possible, there's nothing you can learn to predict
- **B.** Learning one hypothesis hurts learning other hypotheses
- **C.** If someone gives you lunch for free, they'll expect something back
- **D.** Generalizing to new stimuli can hurt a learner

Poll — answer

A. When all hypotheses are possible, there's nothing you can learn to predict.

A flat prior over every possible world means data carries no leverage — every continuation stays 50/50. To learn, you must *commit* to some worlds being more likely than others. **That commitment is your prior — and it is why your prior matters.**

Where we are

Welcome + Clusters walkthrough	0:00
The generalization problem	0:10
The Bayesian generalization framework + size principle	0:18
Rectangle game + number game	0:40
Break	1:08
Student presentation — Shohei	1:15
No Free Lunch	1:40
Hierarchical Bayes + close	1:50

Hierarchical Bayes

Back to Chibany's class

Every student in Chibany's class has their **own** tonkatsu-vs-hamburger rate θ_i .

The rates aren't identical — but they aren't unrelated either. They're all Chibany's customers.

Should learning Aoi's rate tell us anything about Ben's?

Priors over priors

Put a prior on the **parameters of the prior** — and treat (a, b) themselves as unknown:

$$\theta_i \sim \text{Beta}(a, b)$$

The two-level model: $(a, b) \longrightarrow \theta_i \longrightarrow$ observed bentos.

(a, b) is the *shared* structure; each θ_i is a student.

Why it matters

A hierarchical model lets a learner **learn the prior from data** — exactly the inductive bias No Free Lunch said you cannot do without.

- (a, b) is learned from *all* the students together
- It is how “overhypotheses” get acquired — the shape bias, object vs. substance kinds (Kemp, Perfors & Tenenbaum, 2007)

Three ways to pool the data

Six students, six bento records. How do you estimate their rates?

Approach	Model	Problem
Complete pooling	one shared θ for everyone	ignores that students differ
No pooling	a separate θ_i , all unrelated	ignores that they're all Chibany's customers
Hierarchical	$\theta_i \sim \text{Beta}(a, b)$	<i>the middle path</i> — borrows strength

Hierarchical Bayes = **partial pooling**.

The two-level model

Built up, top to bottom:

$$(a, b) \sim \text{prior}$$

$$\theta_i \mid a, b \sim \text{Beta}(a, b)$$

$$k_i \mid \theta_i \sim \text{Binomial}(n_i, \theta_i)$$

k_i — tonkatsu count for student i . n_i — that student's total bentos.

Inference — no closed form

We want the posterior over *everything*:

$$p(a, b, \{\theta_i\} \mid \text{data})$$

Unlike the Beta-Binomial of Week 3, this has **no clean closed form**.

This is where sampling comes in — and where **GenJAX** earns its place (the hierarchical `bento_day()` exercise builds exactly this model).

Shrinkage — borrowing strength

For each student, compare:

- their **raw** tonkatsu fraction k_i/n_i
- their **posterior mean** $\hat{\theta}_i$

The posterior means are pulled **toward the group mean** — *a lot* when a student has little data, barely at all when they have a lot.

This is hierarchical Bayes **automatically borrowing strength** across students.

Overhypotheses — learning the bias

With the hierarchy in hand, **overhypotheses** become precise.

The shape bias; the object-vs-substance distinction — these are *second-level* hypotheses, learned as a distribution over *kinds of concept* (Kemp, Perfors & Tenenbaum, 2007).

No Free Lunch said a learner needs inductive bias. The hierarchy is where a learner **acquires** it — instead of being born with it.

Close

Before next week

Read **T3 Ch 5 — Mixture models** before Week 5.

It formalizes the same partial-pooling idea you just saw, and feeds directly into **Problem 3** of the Clusters assignment.

Clusters is due **Fri Jun 5, 8:00 PM**.

Next week — Week 5

Bayes nets + causal Bayes nets.

From “concepts as sets” to **structured probabilistic models** — graphs that encode how variables depend on each other, and what happens when you *intervene*.

Check the readings page for the Week 5 paper + presenter.